

# L-functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

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## Abstract

We give a single manifestly finite, regulator-free functional realizing in one object L-functions quasi-periods and period polynomials of meromorphic modular forms on  $\mathrm{SL}_2(\mathbb{Z})$ , with generic poles throughout the upper half-plane  $\mathbb{H}$  and at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$ . Additionally, our construction preserves holomorphicity. It is a two-segment contour integral,  $L^*(f, s) = L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S)$ , where  $L_T^*$  and  $L_S^*$  have distinct integration kernels, and  $\gamma_{\tau_0}^S$  is a path in  $\mathbb{H}$  from  $S(\tau_0)$  to  $\tau_0$ , and the same for  $\gamma_{\tau_0}^T$ . For weakly holomorphic forms our L-function is independent of  $\tau_0$  and of the path, and reproduces classical L-functions for holomorphic modular forms, reproduces the L-functions of weak modular forms in [1], which previously needed non-holomorphic regularization to be defined, and reproduces Brown's quasi-periods [2]. For properly meromorphic modular forms with poles inside the interior of  $\mathbb{H}$ , our L-function depends on the homotopy classes, defined in terms of pole-crossings. There is a natural choice of trivial homotopy class, which renders our L-functions canonical when the modular form has a pole in the interior of the fundamental domain, generalizing [3].

## 1 Introduction and preliminary concepts

Our primary goal in this note is to put forth a unifying generalization of the L-function associated to a given modular form. Our L-function naturally encompasses the classical case of L-functions associated to holomorphic modular forms [4], and seamlessly unifies Ref. [1]'s extension to L-functions of modular forms with poles at the cusp with Ref. [3]'s extension to L-functions of modular forms with poles at finite locations, exclusively away from cusps. Secondarily, we show that our L-functions reproduce the numerical results for quasi-periods associated to weak cusp forms [2].

*Note:* Throughout this document, we refer to L-integrals and L-functions. The classical L-function-as-Dirichlet-series which has been suitably extended to the complex  $s$  plane is simply not defined when the modular form that defines the L-function has poles either at the cusp or at finite locations within the upper half-plane. So, if there is ever a doubt, we state clearly: *any L-function in this note is defined by an L-integral*. The focus of this note is to define L-integrals for any integer weight  $\mathrm{SL}_2(\mathbb{Z})$  modular form which may have poles at cusps or away from cusps.

### 1.1 Modular forms

Write  $\mathrm{Re}(z)$  and  $\mathrm{Im}(z)$  for the real and imaginary parts of a complex number  $z \in \mathbb{C}$ . Let  $\mathbb{H} := \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\}$  denote the upper half-plane.  $\mathrm{SL}_2(\mathbb{Z})$  is the group generated by  $T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ .

There is a natural  $\mathrm{SL}_2(\mathbb{Z})$  action on functions  $f : \mathbb{H} \rightarrow \mathbb{C}$ , graded by integers  $k$ :  $(f|_k \gamma)(\tau) := (c\tau + d)^{-k} f(\gamma \cdot \tau)$  for all  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ . Similarly,  $\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d}$ .

In this vein, a *holomorphic modular form* of weight  $k$  for  $\mathrm{SL}_2(\mathbb{Z})$  is a holomorphic function  $f : \mathbb{H} \rightarrow \mathbb{C}$  such that  $(f|_k \gamma)(\tau) = f(\tau)$  for all  $\gamma \in \mathrm{SL}_2(\mathbb{Z})$  (aka  $f(\tau + 1) = f(-\frac{1}{\tau})\tau^{-k} = f(\tau)$ ) for all

$\tau \in \mathbb{H}$ , and  $f(\tau) = \mathcal{O}(1)$  as  $\text{Im}(\tau) \rightarrow \infty$ , and which has no poles at finite locations within  $\mathbb{H}$ . By virtue of the periodicity  $f(\tau) = f(\tau + 1)$  and holomorphicity, each holomorphic modular form has a  $q$ -series decomposition,

$$f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau} = \sum_n c_f(n) q^n \quad , \quad q := e^{2\pi i \tau} \quad . \quad (1)$$

In order for  $f$  to be bounded as  $\tau \rightarrow i\infty$ , we conclude that  $c_f(n) = 0$  for all  $n < 0$ , i.e. that  $f = \sum_{n \geq 0} c_f(n) q^n$ ; for such modular forms  $f$  we write  $f \in M_k$ . If a modular form  $f$  additionally has a vanishing constant term, i.e. if  $c_f(0) = 0$ , then  $f$  is a *cusp form*, and we write  $f \in S_k$ . Clearly,  $S_k \subset M_k$ . Further,  $M_k$  is a finite-dimensional vector-space, as is  $S_k$ . Holomorphic  $\text{SL}_2(\mathbb{Z})$  modular forms must have even integer weights, and  $k \geq 4$  [4].

A product of modular forms of weights  $k_1$  and  $k_2$  is a modular form of weight  $k_1 + k_2$ , so  $M_* := \bigoplus_k M_k$  is naturally a (graded) ring. This graded ring is freely generated by the two modular forms  $E_4 \in M_4$  and  $E_6 \in M_6$ , the  $k = 4, 6$  specializations of the *weight  $k$  Eisenstein series*  $E_k$ , which may be defined as follows:

$$E_k := \frac{1}{2} \sum_{(m,n)=1} \frac{1}{(m+n\tau)^k} = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \quad , \quad (2)$$

where  $\sum_{(m,n)=1}$  denotes summing over all relatively prime  $(m, n) \in \mathbb{Z}^2$ , the Bernoulli number  $B_n$  is the coefficient of  $t^n/n!$  of the Laurent expansion of  $t/(e^t - 1)$  about  $t = 0$ , and  $\sigma_m(n) := \sum_{d|n} d^m$  is the Divisor function where  $d|n$  means  $n/d \in \mathbb{N}$ . In short,  $M_* = \mathbb{C}[E_4, E_6]$ . We now define the *modular discriminant*:

$$\Delta := \frac{E_4^3 - E_6^2}{1728} = q + \mathcal{O}(q^2) \in S_{12} \quad . \quad (3)$$

Holomorphicity is most canonically relaxed via allowing a modular form to have a pole at the cusp of  $\text{SL}_2(\mathbb{Z})$ , i.e. for a weight  $k$  modular form for  $\text{SL}_2(\mathbb{Z})$  to have negative powers of  $q$  in its  $q$ -series expansion. In this case,  $f$  diverges as  $\tau \rightarrow i\infty$  as  $f \sim e^{-2\pi N \tau}$ , where  $-N$  is the largest negative number such that  $c_f(n)$  is non-zero. Such a modular form  $f$  is called a *weak modular form*, and we write  $f \in M_k^!$ . If a weak modular form  $f$  has a vanishing coefficient of  $q^0$ , i.e. if  $c_f(0) = 0$ , it is called a *weak cusp form* and we write  $f \in S_k^!$ . Clearly,  $S_k^! \subset M_k^!$ ,  $S_k \subset S_k^!$  and  $M_k \subset M_k^!$ . Two important weak cusp forms in this note are as follows:

$$J := \frac{E_4^3}{\Delta} - 744 = \frac{1}{q} + \mathcal{O}(q) \in S_0^! \quad , \quad (4)$$

$$\widehat{\Delta} = (J^2 + 24J - 393444)\Delta := \frac{1}{q} + \mathcal{O}(q^2) \in S_{12}^! \quad . \quad (5)$$

In this work, we further relax from even weakly holomorphic modular forms, and consider *meromorphic modular forms* of weight  $k$  which are quotients:  $f = \frac{f_1}{f_2}$  where  $f_i \in M_{k_i}$ , the denominator  $f_2$  is not identically zero, and  $k = k_1 - k_2$ . We write  $F_k$  for the vector space of meromorphic modular forms of weight  $k$ , and set  $F_* := \bigoplus_k F_k$ . Then  $F_k$  vanishes unless  $k$  is an even integer, and  $F_*$  may be regarded as a “graded field of fractions” of  $M_*$ .

## 1.2 L-functions for holomorphic modular forms and periods for cusp forms

For any  $f \in S_k$ , one may define the associated L-function:

$$L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s) \stackrel{f \in S_k}{:=} \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty f(it) t^{s-1} dt = \sum_{n \geq 1} \frac{c_f(n)}{n^s} \quad . \quad (6)$$

A similar definition applies for  $f \in M_k$ . (The classical definition of  $L^*(f, s)$  for  $f \in M_k$  is the focus of Remark 3.6, below.) Similarly, one may define a period polynomial for  $f \in S_k$  in terms of this L-function,

$$r_f(X, Y) := (2\pi i)^{k-1} \sum_{\ell=0}^{k-2} (-1)^\ell \binom{k-2}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{k-2-\ell} Y^\ell. \quad (7)$$

This polynomial,  $r_f(X, Y) \in V_{k-2} := \mathbb{C}[X, Y]_{k-2}$ , has a natural action under  $\mathrm{SL}_2(\mathbb{Z})$ ,  $(P|_\gamma)(X, Y) = P(aX + bY, cX + dY)$  for  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ . This period polynomial can be decomposed into a basis of highly constrained polynomials, which are annihilated by  $|(1+T)$  and  $|(1+U+U^2)[6]$ , where  $U := TS$  generates an order-3 cyclic subgroup of  $\mathrm{SL}_2(\mathbb{Z})$ . We define the *periods* of a cusp form to be coefficients of  $r_f$  within this highly constrained sub-space  $W$  of  $V_{k-2}$ , i.e.  $W \subset V_{k-2}$ .

**Why this work?** As classically computed, these L-functions and associated period polynomials (when defined in terms of the special values of the associated L-function) are not defined when  $f$  has poles either at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$  or within the interior of  $\mathbb{H}$ . As stated above, there are extant, but partial and non-overlapping, extensions of the definition of  $L(f, s)$  to  $f \in M_k^!$  [1] or  $f \in F_k$  when  $f$  is regular at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$  [3]. In this note, we provide a construction that unifies these disparate extensions. We start with generic L-integrals in section 2, then specialize to  $f \in M_k^!$  in section 3, and finally explore the more general context of  $f \in F_k$  in section 4.

## 2 A finite L-integral

**Definition 2.1.** Let  $\tau_0 \in \mathbb{H}$  and  $f \in F_k$ , let  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ,  $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  denote the generators of  $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ , and let  $f^{-1}(\infty)$  be the locus of all poles of  $f \in \mathbb{H}$  (inclusive of the cusp of  $\mathrm{SL}_2(\mathbb{Z})$ ). The  $S$ - and  $T$ -segments at  $\tau_0$  are smooth paths in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$  from  $\sigma^{-1}\tau_0$  to  $\tau_0$ ,

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0. \quad (8)$$

Avoiding poles is automatic for  $f \in M_k^!$ , as long as the base-point  $\tau_0$ , is not at a cusp of  $\mathrm{SL}_2(\mathbb{Z})$ , i.e. there is only the trivial homotopy class. When  $f \in F_k$ , and  $f$  has poles at finite and non-zero imaginary heights, integration contours fall into nontrivial homotopy classes.

**Definition 2.2.** Let  $\zeta(s, x) = \sum_{m \geq 0} (m+x)^{-s}$  as a function  $\zeta : \mathbb{C}_s \times (\mathbb{C} \setminus \mathbb{N})_x \rightarrow \mathbb{C}$  denote the Hurwitz zeta function (meromorphic in  $s$ , holomorphic in  $x \in \mathbb{H}$ ). We define Hurwitz  $T$ -kernel as

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(1-(k-s), \tau+1). \quad (9)$$

**Definition 2.3** (Generic L-integral). Let  $f \in F_k$ ,  $\tau_0 \in \mathbb{H}$ , and  $s \in \mathbb{C}$ . Further, let  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$  be the  $S$ - and  $T$ -segments at  $\tau_0$ , as in Definition 2.1, and let  $k_T$  be as defined in Definition 2.2. The L-integral of  $f$  is

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) := e^{-\pi i s/2} \left( \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right) \quad (10)$$

We use this  $L$ -integral to define a generalized  $L$ -function:

**Definition 2.4** (Generic L-function). Let  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  be as defined in Definition 2.3. We define the generalized L-function of  $f \in F_k$  to be,

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (11)$$

Having defined the L-integral and L-function, we prove they are manifestly finite for  $f \in F_k$ :

**Lemma 2.5** (Finiteness of L-integrals). *For  $f \in F_k$ , both  $L_S^*(f, s; \gamma_{\tau_0}^S)$  and  $L_T^*(f, s; \gamma_{\tau_0}^T)$  in Definition 2.3 are each, separately finite for every  $s \in \mathbb{C}$ , every  $\tau_0 \in \mathbb{H}$ , and for every path  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ .*

*Proof.* Both segments are compact and lie in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , so  $f$  is continuous on them;  $\tau^{s-1}$  is continuous on  $\mathbb{H}$ , and  $\tilde{k}_T(\tau, s)$  is continuous there since  $\zeta(\nu, \tau + 1)$  is holomorphic in  $\tau \in \mathbb{H}$  (Definition 2.2). A continuous integrand on a compact path is integrable, for every  $s \in \mathbb{C}$ .  $\square$

We now turn to path-dependence and  $\tau_0$ -independence of the L-integral and the L-function of  $f \in F_k$ . This holds for all  $f \in M_k^!$ :

**Proposition 2.6** (Basepoint & path independence for  $f \in M_k^!$ ). *For  $f \in M_k^!$  the value  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is independent of  $\tau_0 \in \mathbb{H}$  and of the paths: with no interior poles,  $\mathbb{H}$  is simply connected, so all admissible contours are homotopic. We write  $L^*(f, s)$  for the common value and  $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$ .*

*Proof.* With no interior poles the integrands  $f(\tau)\tau^{s-1}$  and  $f(\tau)\tilde{k}_T(\tau, s)$  are holomorphic on  $\mathbb{H}$ , which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of  $\tau_0$  show that variations in  $\tau_0$  for the S-term and T-term in  $L^*(f, s)$  cancel, i.e. that  $\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = -\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T)$ . For  $L_T^*$ , note that the  $T$ -segment runs from  $T^{-1}\tau_0 = \tau_0 - 1$  to  $\tau_0$  — two endpoints a unit apart, moving together with  $\tau_0$  — so by Leibniz's rule and the  $T$ -periodicity  $f(\tau_0 - 1) = f(\tau_0)$ ,

$$\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s), \quad (12)$$

where  $\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s)$ . This is the whole role of  $\delta_T$ : the unit gap between the endpoints turns the  $\partial_{\tau_0}$  boundary term into a unit shift of the kernel. That shift is elementary — the kernel was built for it — since  $\zeta(\nu, x + 1) - \zeta(\nu, x) = -x^{-\nu}$  gives, on both terms at  $x = \tau + 1$ ,

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s} \quad (13)$$

$$\implies \partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (14)$$

The  $S$ -segment runs  $-1/\tau_0 \rightarrow \tau_0$ ; its lower endpoint moves with  $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$ , so by Leibniz, the modularity  $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ , and the phase  $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)}\tau_0^{1-s}$ ,

$$\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = f(\tau_0)\tau_0^{s-1} - f(-1/\tau_0)(-1/\tau_0)^{s-1}\tau_0^{-2} = f(\tau_0)[\tau_0^{s-1} - e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (15)$$

Thus  $\tau_0$ -variations are equal and opposite,  $\partial_{\tau_0} L_S^* = -\partial_{\tau_0} L_T^*$ . Thus  $\partial_{\tau_0} L^* = \partial_{\tau_0} L_S^* + \partial_{\tau_0} L_T^* = 0$ .  $\square$

**Remark 2.7** (Recovering the classical L-integral for  $f \in S_k$ ). *Let  $f \in S_k$ . As  $f^{-1}(\infty)$  does not include the cusp  $i\infty$ ,  $\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  exists. In this limit, the  $S$ -segment maps onto the standard  $L$ -function contour,  $[0, i\infty]$ . Simultaneously, as  $f(it) \sim e^{-2\pi t}$  as  $t \rightarrow \infty$ , the  $T$ -segment vanishes exponentially. Thus, in this limit,  $L^*(f, s)$  exactly maps onto the classical  $L$ -function integral [4]:*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt. \quad (16)$$

### 3 L-functions for weak modular forms

**Proposition 3.1.** Let  $\Gamma(s, x) := \int_x^\infty e^{-t} t^{s-1} dt$  denote the incomplete Gamma function. For  $n \neq 0$ ,  $\nu \in \mathbb{C}$ , and  $z \in \mathbb{H}$ ,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (17)$$

*Proof.* Write  $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$  and shift  $u \mapsto u+m$  in the  $m$ -th term; as  $e^{2\pi i n(u+m)} = e^{2\pi i n u}$  for integer  $m$ , the unit segments stack into one horizontal ray,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{z+m}^{z+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (18)$$

Substitute  $w = -2\pi i n v$ , so that  $v = iw/(2\pi n)$ ,  $dv = i dw/(2\pi n)$ , and  $e^{2\pi i n v} = e^{-w}$ : the lower endpoint becomes  $w = -2\pi i n z$ , and for  $n > 0$  the ray rotates to  $w \rightarrow +\infty$  along the positive reals — legal, since  $e^{-w} w^{-\nu}$  decays in the right half-plane, so the closing arc vanishes. With  $v^{-\nu} = i^{-\nu} (2\pi n)^{\nu-\nu} w^{-\nu}$ ,

$$\int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv = i^{1-\nu} (2\pi n)^{\nu-1} \int_{-2\pi i n z}^{\infty} e^{-w} w^{-\nu} dw = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (19)$$

For  $n < 0$  the ray instead grows as  $\text{Im} v \rightarrow \infty$  and the rotation is illegal; but both sides are analytic in  $n$  and agree for  $n > 0$ , so the identity continues to  $n < 0$  with  $\Gamma$  on its principal branch. At  $n = 0$  the prefactor  $(2\pi n)^{\nu-1}$  is singular; that case is handled separately.  $\square$

**Theorem 3.2** (The L-function for  $f \in M_k^!$ ). For  $f \in M_k^!$ , we may set  $\tau_0 = i$ , and prove

$$L^*(f, s) = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right). \quad (20)$$

Here, trivially,  $L^*(f, s) = i^k L^*(f, k-s)$ . The L-function is  $L(f, s) := (2\pi)^s L^*(f, s)/\Gamma(s)$ .

*Proof.* By path independence and base-point independence from Proposition 2.6, we may freely set  $\tau_0 = i$ . Here,  $L_S^*(f, s; \gamma_{\tau_0}^S) = 0$  (the  $S$ -contour vanishes in length and there are no nontrivial homotopy classes as  $f \in M_k^!$  has no poles which a finite contour can wrap), and the L-integral simplifies dramatically:

$$L^*(f, s) = L_T^*(f, s; \gamma_i^T) = e^{-\pi i s/2} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau. \quad (21)$$

We then expand  $f$  in terms of its q-series,  $\sum_n c_f(n) e^{2\pi i n \tau}$ , and substitute the  $u$ -variable of Proposition 3.1 as  $u = \tau + 1$ , and note that  $e^{2\pi i n(u-1)} = e^{2\pi i n u}$ . When  $n \neq 0$ , straightforward substitutions of  $\nu, z$  into Proposition 3.1 suffice:  $1-\nu = s$  gives the contribution of the first term in  $\tilde{k}_T(\tau, s)$ , and  $1-\nu = k-s$  gives the second:

$$e^{-\pi i s/2} \left( e^{\pi i s/2} \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} - e^{i\pi(s-1)} e^{\pi i(k-s)/2} \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (22)$$

We may then evaluate the  $n = 0$  mode's contribution follows from noting the closed form solution for the definite integral,

$$\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma-1}. \quad (23)$$

This result follows from noting  $\partial_a(\zeta(\sigma-1, a)/(1-\sigma)) = \zeta(\sigma, a)$  and  $\zeta(\sigma-1, a+1) - \zeta(\sigma-1, a) = -a^{1-\sigma}$ . Applying this to the two distinct terms of  $\tilde{k}_T(\tau, s)$  at  $\sigma = 1-s$  and  $\sigma = 1-(k-s)$  gives

$$e^{-\pi is/2} \int_{i-1}^i c_f(0) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right). \quad (24)$$

Combining the  $n=0$  contributions with the  $n \neq 0$  contributions reproduces Eq. (20). Finally, the functional equation,  $L^*(f, s) = i^k L^*(f, k-s)$ , follows by direct inspection.  $\square$

### 3.1 Explicit S- and T-contours for general $\tau_0$

**Lemma 3.3** ( $L_T^*$  for  $M_k^!$ ). *For  $f \in M_k^!$ , and  $\tau_0 \in \mathbb{H}$  away from cusps,  $L_T^*(f, s; \gamma_{\tau_0}^T)$  evaluates to:*

$$L_T^*(f, s; \gamma_{\tau_0}^T) = -c_f(0) \left( \frac{(\tau_0/i)^s}{s} + i^k \frac{(\tau_0/i)^{k-s}}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (25)$$

*Proof.* Expand  $f = \sum_n c_f(n) q^n$  on the compact segment  $[\tau_0-1, \tau_0]$ , and restrict attention to the sub-sum,  $\sum_{n \neq 0}$ . For each such  $n \neq 0$ , apply the Proposition 3.1 in  $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$  at  $\nu = 1-s$  and  $\nu = s-(k-1)$ . The phases for the second term work-out identically as in the proof of Theorem 3.2.

For  $n=0$ , the constant  $c_f(0)$  multiplies  $e^{-\pi is/2} \int_{\tau_0-1}^{\tau_0} \tilde{k}_T(\tau, s) d\tau$ ; the Hurwitz antiderivative  $\int_{\tau_0-1}^{\tau_0} \zeta(\sigma, \tau+1) d\tau = \tau_0^{1-\sigma}/(\sigma-1) -$  from  $\partial_\tau \zeta(\sigma-1, \tau+1) = -(\sigma-1)\zeta(\sigma, \tau+1)$  and the shift  $\zeta(\sigma-1, \tau_0+1) - \zeta(\sigma-1, \tau_0) = -\tau_0^{1-\sigma}$  — applied to the two halves of  $\tilde{k}_T$  at  $\sigma = 1-s$  and  $\sigma = s-k+1$  yields  $-e^{-\pi is/2} \tau_0^s/s = -(\tau_0/i)^s/s$  and  $-e^{\pi is/2} \tau_0^{k-s}/(k-s) = -i^k (\tau_0/i)^{k-s}/(k-s)$  respectively.  $\square$

**Corollary 3.4** ( $L_T^*$  as  $\tau_0 \rightarrow i\infty$ ). *Let  $f \in S_k^!$ , where  $f = \sum_{n \geq -N} c_f(n) q^n$  for some  $-N \in \mathbb{Z}$ . The T-contour's contribution to  $L^*(f, s)$  is given by Lemma 3.3. As  $\tau_0 = it$  goes to the cusp  $i\infty$ , the T-contour scales exactly as  $f(it)$  scales:  $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(it) \sim e^{2\pi Nt}$ .*

*Proof.* For the asymptotics,  $\Gamma(\sigma, z) \sim z^{\sigma-1} e^{-z}$  as  $|z| \rightarrow \infty$  for  $|\arg z| < \frac{3\pi}{2}$  [10, §8.11] gives  $|c_f(n)|$ -weighted go as  $e^{-2\pi n \text{Im}(\tau_0)}$ . The smallest mode  $n$  in the q-series, i.e.  $n = -N$ , dominates the scaling of  $f(it)$  and of  $L_T^*(f, s; \gamma_{it}^T)$ . Combining these results yields the claimed asymptotics.  $\square$

Now, exploiting  $\tau_0$ -independence of  $L^*(f, s)$ , its explicit value as given in Theorem 3.2, and the explicit T-contribution  $L_T^*(f, s; \gamma_{\tau_0}^T)$  trivially implies clean, closed-form expressions for  $L_S^*(f, s; \gamma_{\tau_0}^S)$ :

**Corollary 3.5** ( $L_S^*$  for  $M_k^!$ ). *For  $f \in M_k^!$  and  $\tau_0 \in \mathbb{H}$ ,  $L_S^*(f, s; \gamma_{\tau_0}^S)$  evaluates to*

$$L_S^*(f, s; \gamma_{\tau_0}^S) = c_f(0) \left( \frac{(\tau_0/i)^s - 1}{s} + i^k \frac{(\tau_0/i)^{k-s} - 1}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, 2\pi n) - \Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n) - \Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (26)$$

*Proof.* Recall  $L^*(f, s) - L_T^*(f, s; \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S)$ . Now,  $L^*(f, s)$  for  $f \in M_k^!$  is given by Theorem 3.2, and  $L_T^*(f, s; \gamma_{\tau_0}^T)$  is given by Lemma 3.3. Their difference yields Eq. (26).  $\square$

### 3.2 Classical Mellin recovery and canonical uniqueness.

As discussed in Remark 2.7, for  $f \in S_k$ , taking the cusp-limit  $\tau_0 \rightarrow i\infty$  in Theorem 3.2 collapses the  $S$ -segment onto the classical Mellin contour  $[0, i\infty]$  while via Corollary 3.4 the  $T$ -segment vanishes exponentially as  $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(\tau_0) \sim e^{-2\pi\text{Im}(\tau_0)N}$  (as  $f \in S_k \Leftrightarrow f = \sum_{n \geq N} c_f(n)q^n$  with  $N \geq 1$ ). I.e., here, one recovers the classical L-integral of  $f$  without obstruction:

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt. \quad (27)$$

This story changes when  $f = 1/q^N(1 + O(q)) \in S_k^!$  for some  $N \geq 1$ .

In short, when  $f$  has a polar  $q$ -series, it has a pole at the cusp of  $\text{SL}_2(\mathbb{Z})$ . Thus, the limit  $\tau_0 \rightarrow i\infty$ , which was crucial to recovering the classical Mellin contour in Remark 2.7 simply does not exist.

However, it can still be instructive to examine  $L_T^*(f, s; \gamma_{\tau_0}^T)$ , and  $L_S^*(f, s; \gamma_{\tau_0}^S)$  as  $\tau_0$  goes towards  $i\infty$ . In this context, while of course the sum  $L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S) = L^*(f, s)$  is manifestly invariant under changes of the base-point, each of these two terms can indeed change values. Indeed, Corollary 3.4 clearly shows that as  $\tau_0 \rightarrow i\infty$ , when  $f \in S_k^!$  has negative powers of  $q$  in its  $q$ -series, then  $L_T^*(f, s; \gamma_{\tau_0}^T) \sim e^{2\pi N \text{Im}(\tau_0)}$ . Discarding  $\gamma_{\tau_0}^T$  therefore does violence to the result: When the  $S$ -contour is isolated from its  $T$ -partner, and  $\tau_0 \rightarrow i\infty$ , it diverges. This is the precise reason that [1] needed to regularize their divergent L-integral representation for properly weak cusp forms.

That said, there is one further aspect of our L-function even in the classical case of  $f \in M_k$  that is worthy of remark:

**Remark 3.6** (Recovering the classical L-integral for  $f \in M_k$ ). *In textbook treatments [4], when  $f \in M_k$ , its L-integral is defined as*

$$\begin{aligned} L^*(f, s)|_{\text{cl}} &:= \int_0^\infty (f(it) - c_f(0))t^{s-1}dt = \frac{\Gamma(s)}{(2\pi)^s} \sum_{n=1}^\infty \frac{c_f(n)}{n^s} \\ &= -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n=1}^\infty c_f(n) \left( \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) \end{aligned} \quad (28)$$

*Classically, this constant-term subtraction is needed because  $\int_0^\infty t^{s-1}dt$  is a pure divergence, so the  $q$ -series cannot commute with the integration contour.*

*However, as is clear from our general Theorem 3.2 for L-integrals for  $f \in M_k^!$ , and from the specific results of the  $S$ - and  $T$ -contours in Lemma 3.3 and Corollary 3.5, our L-integral requires no such explicit subtraction of the constant term  $-c_f(0)$ . Thus, not only does our prescription naturally handle divergences at poles when  $f \in M_k^!$  has polar terms in its  $q$ -series, but it also obviates the need to by-hand remove the constant mode from the kernel of the L-integral that is needed when defining L-functions in terms of the classical Mellin contour,  $\int_0^\infty$ .*

### 3.3 Period polynomials, periods and quasi-periods

Because  $L^*(f, s)$  is canonical for every  $f \in S_k^!$ , and because period polynomials for a given modular form can be written in terms of their L-functions evaluated at integer- $s$  in the critical strip, they need no separate construction.

**Definition 3.7** (Period polynomial for  $S_k^!$ ). *For  $f \in S_k^!$  and  $n = k - 2$ , the period polynomial is*

$$r_f(X, Y) := (2\pi i)^{n+1} \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^\ell. \quad (29)$$

**Proposition 3.8** (Period polynomial as a contour integral). *Let  $n := k - 2$ . With  $(X - \tau Y)^n$  in place of the  $L$ -weight  $\tau^{s-1}$ , and  $\mathbf{K}_T(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}_T(\tau, \ell + 1) X^{n-\ell} Y^\ell$  in place of  $\tilde{k}_T$ , the same two segments assemble  $r_f$  directly:*

$$r_f(X, Y) = (2\pi i)^{n+1} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) (X - \tau Y)^n d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau \right]. \quad (30)$$

*Proof.* By Definition 2.3, the  $X^{n-\ell} Y^\ell$  coefficient of the bracket is

$$r_f|_{X^{n-\ell} Y^\ell} = (-1)^\ell \binom{n}{\ell} i^{\ell+1} [L_S^*(f, \ell + 1) + L_T^*(f, \ell + 1)] = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell + 1). \quad (31)$$

This matches Definition 3.7 term by term, and proves the claim.  $\square$

Having now defined the period polynomial associated to a weak cusp form  $f \in S_k^!$ , we now seek to find its (*quasi*-)periods. First, a number of helpful definitions. For starters, let  $f \in S_k^!$ ,  $\tau_0 \in \mathbb{H}$  and  $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ , and define

$$C_\gamma(f; \tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f, \quad \text{where } \omega_f := f(\tau)(X - \tau Y)^n d\tau. \quad (32)$$

Finally, it is helpful to define two slight abstractions related to  $\mathbf{K}_T(\tau; X, Y)$  and to  $L_T^*(f, s; \gamma_{\tau_0}^T)$ :

$$\chi(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \zeta(-\ell, \tau + 1) X^{n-\ell} Y^\ell, \quad (33)$$

$$E(f; \tau_0) := (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \chi(\tau; X, Y) d\tau. \quad (34)$$

For each  $\tau$ ,  $\chi(\tau; \cdot, \cdot) \in V_{k-2} = \mathbb{C}[X, Y]_{k-2}$ ; the  $\mathrm{SL}_2(\mathbb{Z})$ -slash  $(P|_\gamma)(X, Y) := P(aX + bY, cX + dY)$  acts on the polynomial variables alone, holding  $\tau$  fixed:

$$\chi|_\gamma(\tau; X, Y) := \chi(\tau; aX + bY, cX + dY), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}). \quad (35)$$

Thus  $|_T$  sends  $(X, Y) \mapsto (X + Y, Y)$  with  $\tau$  untouched. Being linear and  $\tau$ -independent,  $|_\gamma$  commutes with  $\int d\tau$ , so slashing the polynomial  $E$  agrees with slashing  $\chi$  under the integral — the two occurrences of  $|_\gamma$  in Lemma 3.9 denote the same operation.

**Lemma 3.9.** *Let  $f \in S_k^!$ ,  $\tau_0 \in \mathbb{H}$ , and let  $\omega_f$ ,  $C_\gamma(f; \tau_0)$ ,  $\chi(\tau; X, Y)$ , and  $E(f; \tau_0)$  be as defined in Eqs. (32), (33) and (34). Then  $\hat{C}_\gamma := C_\gamma - (E|_\gamma - E)$  is a 1-cocycle for  $\mathrm{SL}_2(\mathbb{Z})$  with  $\hat{C}_S = r_f$  and  $\hat{C}_T = 0$ .*

*Proof.* First, we prove  $\hat{C}_\gamma$  is a cocycle. Path-additivity and covariance of  $\omega_f$  give the Eichler–Shimura cocycle relation  $C_{\gamma\delta} = C_\gamma|_\delta + C_\delta$  [11, 6]; as  $\gamma \mapsto E|_\gamma - E$  is a coboundary,  $\hat{C}$  is a cocycle.

Having established it is a cocycle, we now show that  $\hat{C}_S = r_f$ . At  $s = \ell + 1$ , Definition 2.2 reads  $\tilde{k}_T(\tau, \ell + 1) = \zeta(-\ell, \tau + 1) - (-1)^\ell \zeta(-(n - \ell), \tau + 1)$ ; assembling  $\mathbf{K}_T$  and re-indexing  $\ell \mapsto n - \ell$  in the second term (with  $n$  even) gives  $\mathbf{K}_T = \chi - \chi|_S$ . So the  $T$ -term of  $r_f$  is  $E - E|_S$ , and by Proposition 3.8  $r_f = C_S + (E - E|_S) = C_S - (E|_S - E) = \hat{C}_S$ .

Finally, we show that  $\hat{C}_T = 0$ . The Hurwitz shift  $\zeta(-\ell, \tau) = \zeta(-\ell, \tau + 1) + \tau^\ell$  gives the point-wise identity  $\chi(\tau - 1; X, Y) - \chi(\tau; X, Y) = (X - \tau Y)^n$ ; applied twice it shows  $A(\tau) :=$

$\chi(\tau; X+Y, Y) - \chi(\tau-1; X, Y)$  satisfies  $A(\tau-1) = A(\tau)$ . Crucially, because  $A(\tau)$  is a polynomial, and because  $A(\tau) = A(\tau+1)$  for generic  $\tau$ , then  $A(\tau)$  is constant:  $\partial_\tau A(\tau) = 0 \Leftrightarrow A(\tau) = A_0$ . Thus,

$$\begin{aligned}
\hat{C}_T &= C_T - (E|_T - E) \\
&= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau)(X - \tau Y)^n d\tau - (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau)[\chi|_T(\tau; X, Y) - \chi(\tau; X, Y)] d\tau \\
&= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \left( [\chi(\tau-1; X, Y) - \chi(\tau; X, Y)] - [\chi(\tau; X+Y, Y) - \chi(\tau; X, Y)] \right) d\tau \\
&= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) [\chi(\tau-1; X, Y) - \chi(\tau; X+Y, Y)] d\tau = -(2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} f(\tau) A(\tau) d\tau \\
&= -(2\pi i)^{n+1} A_0 \int_{\tau_0-1}^{\tau_0} f(\tau) d\tau = -(2\pi i)^{n+1} A_0 c_f(0) = 0,
\end{aligned} \tag{36}$$

as all elements  $f \in S_k^!$  must have  $c_f(0) = 0$ .  $\square$

**Definition 3.10.**  $W := \{w \in \mathbb{C}[X, Y]_{k-2} : w|_{(1+S)} = w|_{(1+U+U^2)} = 0\} \subset V_n$

**Lemma 3.11.** For  $f \in S_k^!$ , and  $W \subset V_n$  from Definition 3.10, the period polynomial  $r_f(X, Y) \in W$ .

*Proof.* Let  $\hat{C}_\gamma$  be the parabolic cocycle of Lemma 3.9. Thus,  $r_f = \hat{C}_S$  and  $\hat{C}_T = 0$ . Now, note that  $\hat{C}_\gamma(f; \tau_0) = 0$  when  $\gamma = \pm I$  for the simple reason that when  $\gamma = \pm I$ ,  $\hat{C}_\gamma$  is a closed-contour integral of a function which has no poles in  $\mathbb{H}$ . Finally, direct computation, starting from the identities  $-I = U^3 = S^2$  and using the cocycle law  $\hat{C}_{\gamma\delta} = \hat{C}_\gamma|_\delta + \hat{C}_\delta$  and clever use of  $\hat{C}_T = 0$ , shows  $r_f|_{(1+S)} = r_f|_{(1+U+U^2)} = 0$ :

$$\hat{C}_U = \hat{C}_{TS} = \hat{C}_T|_S + \hat{C}_S = \hat{C}_S = r_f, \tag{37}$$

$$0 = \hat{C}_{S^2} = \hat{C}_S|_S + \hat{C}_S = \hat{C}_S|_{(1+S)} = r_f|_{(1+S)}, \tag{38}$$

$$0 = \hat{C}_{U^3} = \hat{C}_U|_{UU} + \hat{C}_{UU} = \hat{C}_U|_{U^2} + (\hat{C}_U|_U + \hat{C}_U) = \hat{C}_U|_{(1+U+U^2)} = r_f|_{(1+U+U^2)}. \tag{39}$$

Thus,  $r_f \in W \subset V_n$ , and finishes the proof.  $\square$

To define periods and quasi-periods, we recall the standard fact that  $\dim(W) = 2 \dim(S_k) + 1$  (see, e.g. [6]), and define the inner-product:

$$\langle P, Q \rangle := \sum_a (-1)^a \binom{n}{a}^{-1} p_a q_{n-a}. \tag{40}$$

Now for the basis elements for  $k$  where  $\dim(S_k) = 1$ :

**Definition 3.12.** Let  $\dim S_k = 1$ ,  $n := k-2$ ,  $p_0(X) := X^n - 1$ ,  $B_\ell := \frac{\ell!}{2\pi i} \oint \frac{t^{-\ell}}{e^t - 1} dt$ , and  $B_m^0(X) := \sum_{i \neq 1} \binom{m}{i} B_i X^{m-i}$ . Motivated by [6, Thm. 1'], we define:

$$W_-^{(k)}(X) := \text{pp} \left( \frac{1}{n-1} (B_{n-1}^0(X) + X^n B_{n-1}^0(\frac{1}{X})) - \frac{1}{3} (B_3^0(X) + X^n B_3^0(\frac{1}{X})) \right), \tag{41}$$

$$W_+^{(k)}(X) := \text{pp} \left( \frac{1}{2} (B_2^0(X) - X^n B_2^0(\frac{1}{X})) + \frac{1}{n} (B_n^0(X) - X^n B_n^0(\frac{1}{X})) - c p_0(X) \right), \tag{42}$$

$$v_0^{(k)}(X) := p_0(X) - W_+^{(k)}(X) \left( \langle W_+^{(k)}, p_0 \rangle / \langle W_+^{(k)}, W_+^{(k)} \rangle \right), \tag{43}$$

where  $\text{pp}$  acts on any polynomial with rational coefficients by multiplying by all coefficient-denominators and dividing by the gcd,  $c$  eliminates the  $p_0(X)$  component of  $W_+^{(k)}$ , and  $\langle P, Q \rangle$  as in Eq. (40).

**Lemma 3.13.** *For  $k$  with  $\dim S_k = 1$ , the space  $W$  of Lemma 3.11 has  $\dim W = 3$ , and  $\{W_+^{(k)}, W_-^{(k)}, v_0^{(k)}\}$  is an orthogonal basis of  $W$  under  $\langle \cdot, \cdot \rangle$  from Eq. (40).*

*Proof.* First, let  $W_\pm^{(k)}(X, Y) := Y^{k-2}W_\pm^{(k)}(X/Y)$ , and let  $v_0^{(k)}(X, Y) := Y^{k-2}v_0^{(k)}(X/Y)$ .

Now, by Eichler–Shimura,  $\dim W = 2 \dim S_k + 1 = 3$ . Theorem 1' of [6] gives the odd and even period polynomials of the cusp form  $R_n \propto \Delta$  as modified-Bernoulli combinations; its  $n = 2$  case is  $r^-(R_2)$ , odd under  $X \mapsto -X$ , and its  $n = 1$  case is  $r^+(R_1)$ , even. Our  $W_-^{(k)}$  is the primitive part of  $r^-(R_2)$ , and  $W_+^{(k)}$  the primitive part of  $r^+(R_1)$  with its Eisenstein  $p_0$ -component removed by  $c$ ; both lie in  $W$ , as does  $p_0 = X^{k-2} - Y^{k-2}$  (even). The three are independent, hence a basis of  $W$ . Since  $k - 2$  is even,  $\langle \cdot, \cdot \rangle$  is symmetric and pairs opposite  $X \mapsto -X$  parities to zero; the odd  $W_-^{(k)}$  is thus orthogonal to the even  $W_+^{(k)}$  and  $v_0^{(k)}$ , while  $v_0^{(k)} \perp W_+^{(k)}$  by construction. Hence  $\{W_+^{(k)}, W_-^{(k)}, v_0^{(k)}\}$  is orthogonal.  $\square$

**Definition 3.14** (Periods and Quasi-Periods). *Let  $k$  be such that  $\dim(S_k) = 1$ ,  $n := k - 2$ ,  $W_\pm^{(k)}(X, Y) := Y^n W_\pm^{(k)}(X/Y)$  be as in Definition 3.12, and  $r_f(X, Y)$  be as in Definition 3.7. Now, define the two numbers  $\omega^\pm(f)$  as follows:*

$$\omega^\pm(f) := \frac{\langle r_f, W_\pm^{(k)} \rangle}{\langle W_\pm^{(k)}, W_\pm^{(k)} \rangle}, \quad (44)$$

where  $\langle \cdot, \cdot \rangle$  is as in Eq. (40). When  $f$  is a cusp form,  $\omega^\pm(f)$  are its periods; when  $f$  is a weak cusp form,  $\omega^\pm(f)$  are its quasi-periods, sometimes denoted  $\eta^\pm(f)$ .

Now, having defined our periods and quasi-periods, we use them together with our L-functions to extract periods and quasi-periods for  $f \in S_k^!$  (when  $\dim(S_k) = 1$ ). To this effect, we recall the notation of [5]. Decomposing the modular weight  $k = 12\lambda + k'$  where  $k' \in \{0, 4, 6, 8, 10, 14\}$ , we see  $\lambda = \dim(S_k)$  when  $\lambda \geq 0$ . For any  $\lambda$ , we may denote  $f_{k,m} := 1/q^m + O(q^{\lambda+1}) \in M_k^!$ . When  $\dim(S_k) = 1$ , we have  $f_{k,m} = 1/q^m + O(q^2)$ . Here, define  $\Delta_k := f_{k,-1} = q + O(q^2) \in S_k$  and  $\hat{\Delta}_k := f_{k,1} = 1/q + O(q^2) \in S_k^!$ . Further, define

$$\text{pd}_k := \begin{pmatrix} \omega^+(\Delta_k) & \omega^-(\Delta_k) \\ \eta^+(\hat{\Delta}_k) & \eta^-(\hat{\Delta}_k) \end{pmatrix}, \quad \alpha_k := \frac{(-1)^{1+k/2} \det(\text{pd}_k)}{(2\pi i)^{k-1} (k-2)!} \quad (45)$$

| $k$ | $\omega^+(\Delta_k)$ | $\omega^-(\Delta_k)/i$ | $\eta^+(\hat{\Delta}_k)$ | $\eta^-(\hat{\Delta}_k)/i$ | $\alpha_k$ |
|-----|----------------------|------------------------|--------------------------|----------------------------|------------|
| 12  | -6.89168e7           | -5.58502e6             | 1.27202e11               | 1.02767e10                 | 1          |
| 16  | -8.81481e11          | -1.54063e10            | 4.14120e13               | 6.83541e11                 | 13/30      |
| 18  | -5.07833e13          | -5.43305e12            | -2.19200e14              | -3.36339e13                | 2/3        |
| 20  | -4.22882e16          | -3.73170e14            | 6.38640e16               | -1.89823e14                | 17/5       |
| 22  | -1.49974e18          | -4.24489e16            | 1.43795e18               | -4.41050e16                | 19/21      |
| 26  | -1.18673e22          | -3.60961e21            | 1.18653e22               | -3.61016e21                | 23/15      |

Table 1: Periods  $\omega^\pm(\Delta_k)$  and quasi-periods  $\eta^\pm(\hat{\Delta}_k)$ , and the rational number  $\alpha_k$ . While  $\alpha_{12} = 1$  matches Brown [2],  $\alpha_k$  for  $k \geq 16$  appear to be new, interesting, rational numbers, albeit with implicit dependence on the period-polynomial basis normalizations—in contrast to the normalization-free Petersson check above. We have not found a clear pattern for the  $\alpha_k$ . See [9] for the scripts that extracted these figures (to 50-digit precision).

Brown–Hain [7, Prop. 5.6] show that  $\det(\text{pd}_k)$  satisfies a Legendre-type relation,  $\det(\text{pd}_k) \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$ . Brown [2, §8.3] numerically finds  $\alpha_k = 1$ ; we reproduce this result at  $k = 12$  and

give the analogous rational multiples for  $k \in \{16, 18, 20, 22, 26\}$  in Table 1, below. Note: The value of  $\alpha_k$  is tied to the normalization of Definition 3.12. Other normalizations, e.g. not including the pp-operator in defining  $W_{\pm}^{(k)}$  would lead to different  $\alpha_k$ -values.

## 4 Meromorphic modular forms

For  $f \in F_k$  with poles in  $\mathbb{H}$ , the two segment integrals of Definition 2.3 depend on their contours; we show their sum depends only on the homotopy class of the pair.

**Proposition 4.1.** *For  $f \in F_k$  and  $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , the L-integral  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  depends only on the homotopy class of its two segments in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , and not on the basepoint  $\tau_0$ . At fixed  $\tau_0$  each of  $L_S^*, L_T^*$  is separately path-independent.*

*Proof.* At fixed  $\tau_0$  the integrands  $f(\tau)\tau^{s-1}$  and  $f(\tau)\tilde{k}_T(\tau, s)$  are holomorphic on  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , so Cauchy's theorem makes each of  $L_S^*, L_T^*$  constant on the homotopy class of its own segment. Under a motion of the basepoint,  $\partial_{\tau_0} L^*$  is evaluated at the moving endpoints alone; the computation of Proposition 2.6 — which uses only  $f(\tau_0 - 1) = f(\tau_0)$  and  $f(-1/\tau_0) = \tau_0^k f(\tau_0)$  — cancels the boundary terms of  $L_S^*$  against those of  $L_T^*$ , the interior poles lying off the contours. Hence  $\partial_{\tau_0} L^* = 0$  along any motion in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , while each summand may separately vary.  $\square$

**Definition 4.2.** *For  $f \in F_k$  and  $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , let  $[\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]$  denote the homotopy classes of the two segments in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ .*

Thus, with  $\gamma_{\tau_0}^S \in [\gamma_{\tau_0}^S]$  and  $\gamma_{\tau_0}^T \in [\gamma_{\tau_0}^T]$ , by Proposition 4.1 the value

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) := L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) \quad (46)$$

is independent of the chosen representative paths and independent of  $\tau_0$  (subject to natural constraints of  $\tau_0$  monodromies around singular points, which move between homotopy classes).

We would like to speak of “the” L-integral. Homotopy classes of course split this into L-integrals for contours in different homotopy classes. And, so, in what follows, we define the reference homotopy classes:

**Definition 4.3** (Reference homotopy class for  $f \in F_k$ ). *Let  $f \in F_k$ , let  $\tau_0$  be the base-point of  $L^*(f, s)$ , and let  $f^{-1}(\infty)$  be the poles of  $f$ . If  $f$  has no poles on the line  $\text{Im}(\tau) = 1$ , choose  $\delta = 0$ . Else, choose  $\delta > 0$  such that no pole lies in the strip  $1 < \text{Im}(\tau) \leq 1 + \delta$ . Set  $\hat{\tau}_0^f := i(1 + \delta)$ . The reference homotopy class  $[\hat{\gamma}_f] = [\hat{\gamma}_f^T] + [\hat{\gamma}_f^S]$  is:*

- $[\hat{\gamma}_f^T]$ : contains the horizontal segment from  $T^{-1}\hat{\tau}_0^f$  to  $\hat{\tau}_0^f$ ;
- $[\hat{\gamma}_f^S]$ : contains the near-vertical path from  $S^{-1}\hat{\tau}_0^f$  to  $\hat{\tau}_0^f$  along the imaginary axis, taking a pole at  $i$  by principal value.

For  $\delta = 0$ , the reference paths match the paths  $\gamma_i^S : i \rightarrow i$  and  $\gamma_i^T : (i-1) \rightarrow i$  for  $L^*(f, s)$  when  $f \in M_k^1$ .

With this definition, we may thus unambiguously write

$$L^*(f, s) := L^*(f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) . \quad (47)$$

Now, having fixed the reference value  $L^*(f, s)$  in the reference homotopy class of Definition 4.3, consider two paths  $\gamma_{\tau_0}^S$  and  $\gamma_{\tau_0}^T$  which may be in any homotopy class. Further, let  $X_S(p), X_T(p) \in \mathbb{Z}$  count the windings  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$  has about each pole  $\tau_p \in f^{-1}(\infty)$  relative to the reference-class.

**Lemma 4.4.** *Let  $f \in F_k$ ,  $r_S(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tau^{s-1})$  and  $r_T(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tilde{k}_T(\tau, s))$ . Then,*

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) = L^*(f, s) + 2\pi i e^{-\pi i s/2} \sum_{\tau_p \in f^{-1}(\infty)} \left[ X_S(\tau_p) r_S(f, \tau_p) + X_T(\tau_p) r_T(f, \tau_p) \right]. \quad (48)$$

*Proof.* Two pairs whose winding data differ by one unit at a single pole  $p$ , on one segment, bound a loop encircling  $p$  once; the residue theorem adds  $2\pi i$  times the residue of that segment's integrand,  $f(\tau)\tau^{s-1}$  for  $\gamma^S$  or  $f(\tau)\tilde{k}_T(\tau, s)$  for  $\gamma^T$ , and the prefactor  $e^{-\pi i s/2}$  of Definition 2.3 multiplies through. Summing over crossings gives (48).  $\square$

This content provides the basics to discuss our finite L-functions for  $f \in F_k$ . However, we wish to present these L-functions in terms of their q-series coefficients, just as we have done for L-functions for  $f \in M_k^!$  in the previous section. To do this, we must introduce polylogarithmic projections of Ref. [3]:

This polylog projection isolates the q-series contribution of a given pole of an  $f \in F_k$  into a known special function, an integral of a polylogarithm of negative index, and leaves behind an object  $(I - \hat{P})f$  with that one pole removed. For our purposes, we find it essential to project-out all poles at, and above, the line  $\text{Im } \tau = 1$ . We must do this because the reference contours which describe our reference homotopy classes are, at a minimum, contour integrals of  $f$  across the horizontal segment from  $i - 1$  to  $i$ . Were we to allow poles at/above this line, q-series diverges on the contour ( $|c_f(n)| \gtrsim n^{m-1} e^{2\pi n \text{Im } \tau_p}$  for a pole of order  $m$  [3]).

Within this general context, we lay-out the starting definitions and facts:

**Definition 4.5.** *Let  $f \in F_k$  have, at a pole  $\tau_p$  of order  $m_p$ , the Laurent principal part  $\sum_{m=1}^{m_p} r_{f, \tau_p}^*(m) (\tau - \tau_p)^{-m}$ ; the pole weights  $r_{f, \tau_p}^*(m)$  are thus its Laurent coefficients. With the polylogarithm  $\text{Li}_s(x) := \sum_{n \geq 1} x^n n^{-s}$ , the polylog projection of  $f$  at  $\tau_p$  is*

$$\hat{P}_{\tau_p} f(\tau) := \sum_{m=1}^{m_p} r_{f, \tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)}), \quad (49)$$

and  $\hat{P} := \sum_{\text{Im } \tau_p \geq 1} \hat{P}_{\tau_p}$ .

**Lemma 4.6.** *The normalized block  $\frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$  has principal part exactly  $(\tau - \tau_p)^{-m}$ , so  $\hat{P}_{\tau_p} f$  matches the principal part of  $f$  and  $(I - \hat{P}_{\tau_p})f$  is holomorphic at  $\tau_p$ .*

*Proof.* By the small- $t$  expansion  $\text{Li}_{1-m}(e^{-t}) = (m-1)! t^{-m} + O(1)$  with  $t = -2\pi i(\tau - \tau_p)$ , the block  $\frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$  has a pure pole  $(\tau - \tau_p)^{-m} + O(1)$  at  $\tau_p$ , with no lower-order poles. Summing against the Laurent coefficients  $r_{f, \tau_p}^*(m)$  therefore reproduces the principal part of  $f$  term by term, leaving  $(I - \hat{P}_{\tau_p})f$  regular at  $\tau_p$ .  $\square$

**Definition 4.7.** *With  $\hat{P}$  as in Definition 4.5, let  $c_{Rf}(n)$  be the Fourier coefficients of the pole-projected form  $(I - \hat{P})f$  in its expansion valid for large  $\text{Im } \tau$ :*

$$(I - \hat{P})f(\tau) = \sum_n c_{Rf}(n) q^n, \quad q = e^{2\pi i \tau}. \quad (50)$$

**Definition 4.8.**  *$F(\infty)$  is the space of meromorphic functions  $g : \mathbb{H} \rightarrow \mathbb{C}$  with  $g(\tau + 1) = g(\tau)$ , finitely many poles in each compact subset of the strip  $|\text{Re } \tau| \leq \frac{1}{2}$ , and bounded pole order at  $q = 0$ .*

**Lemma 4.9.**  $F_k \subset F(\infty)$  and  $(I - \hat{P})f \in F(\infty)$ . Every pole of  $(I - \hat{P})f$  has  $\text{Im } \tau < 1$ , with a maximal height  $y_b < 1$  (set  $y_b = 0$  if  $(I - \hat{P})f$  is holomorphic).

*Proof.* A weakly holomorphic form is  $T$ -periodic, meromorphic, and  $O(e^{C\text{Im}\tau})$  at the cusp, so  $F_k \subset F(\infty)$ ; the projected polylogs  $\text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$  share these properties, so  $(I - \hat{P})f \in F(\infty)$  (and leaves  $F_k$ , being non-modular). By Lemma 4.6,  $\hat{P}$  cancels the principal part of every pole with  $\text{Im}\tau \geq 1$ , so the poles of  $(I - \hat{P})f$  lie strictly below  $\text{Im}\tau = 1$ ; finite in each compact set and accumulating only at  $\text{Im}\tau = 0$ , they have a maximal height  $y_b < 1$ .  $\square$

The Fourier coefficients of a pole-projected form are controlled by its highest pole, an estimate we import from [3]:

**Lemma 4.10.** With  $y_b$  as in Lemma 4.9 and  $m_b$  the maximal order among poles at height  $y_b$ , there is a  $C > 0$  such that, for every  $\varepsilon > 0$ ,

$$C n^{m_b-1} e^{2\pi n y_b} \leq |c_{Rf}(n)| \leq e^{2\pi n(y_b+\varepsilon)}, \quad (51)$$

the upper bound for all large  $n$ , the lower for infinitely many  $n$ . In particular  $|c_{Rf}(n)| e^{-2\pi n} \rightarrow 0$  geometrically, at base  $e^{-2\pi(1-y_b-\varepsilon)}$ .

*Proof.* This is the pole-height/order Fourier-growth estimate of [3] for a  $T$ -periodic meromorphic function, applied to  $(I - \hat{P})f$  (Lemma 4.9).  $\square$

For  $n \neq 0$ , define

$$G_{s,k}(n) := \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (52)$$

Each pole with  $\text{Im } \tau_p \geq 1$  contributes one projected polylog to  $L^*(f, s)$ , whose L-integral we now record.

**Definition 4.11.** For a pole projected at  $z$  with  $\text{Im } z > 1$ , let  $I_N(s, z)$  be the L-integral of the projected polylog  $\text{Li}_{-N}(e^{2\pi i(-z)})$  in the minimal class  $[\hat{\gamma}_f^S], [\hat{\gamma}_f^T]$  of Definition 4.3:

$$I_N(s, z) := L^*(\text{Li}_{-N}(e^{2\pi i(-z)}), s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]). \quad (53)$$

**Lemma 4.12.** Let  $f \in F_k$  have a pole of order  $p$  at  $z$ ,  $\text{Im } z > 1$ , with projection weights  $r_{f,z}^*(m)$  (Definition 4.5). Polylog inversion  $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w) - \delta_{N,0}$  ( $N \in \mathbb{N}_0$ ) resums (53) into (each exponentially convergent, base  $e^{2\pi(1-\text{Im}(z))}$ )

$$L^*(\hat{P}_z f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) = \sum_{m=1}^p r_{f,z}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}(s, z), \quad (54)$$

$$I_N(s, z) = \delta_{N,0} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n=1}^{\infty} n^N e^{2\pi i n z} G_{s,k}(-n). \quad (55)$$

*Proof.* By Proposition 4.1 we may evaluate  $I_N(s, z)$  in the  $\delta \rightarrow 0$  representative of the minimal class; there the projected polylog — regular at  $i$  since  $\text{Im } z > 1$  — has vanishing  $\gamma^S$ -integral and  $\hat{\gamma}_f^T \rightarrow [i-1, i]$ . On  $[i-1, i]$  one has  $|e^{2\pi i(\tau-z)}| > 1$ , so the defining series of  $\text{Li}_{-N}$  diverges; inversion rewrites it as  $(-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n(z-\tau)}$ , convergent since  $|e^{2\pi i(z-\tau)}| < 1$ , and term-by-term integration against  $\tilde{k}_T$  gives  $\sum_n n^N e^{2\pi i n z} G_{s,k}(-n)$ , the mode- $(-n)$  block of Theorem 3.2; for  $N = 0$  the leftover  $-1$  in  $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$  contributes  $\frac{1}{s} + \frac{i^k}{k-s}$ . The rate is  $e^{-2\pi(\text{Im } z - 1)}$ , from  $\Gamma(s, -2\pi n) \sim (2\pi n)^{s-1} e^{2\pi n}$  against  $|e^{2\pi i n z}| = e^{-2\pi n \text{Im } z}$ .  $\square$

Extracting similar results for  $L^*$  evaluated on the polar terms for poles on the line  $\text{Im}(\tau) = 1$  requires more care. To do this away from the special point  $z = i$ , we find it helpful to define the following two functions:

$$\Pi_N(s, z) := \sum_{\lambda=0}^{N+1} \frac{\text{Li}_{\lambda+1-N}(e^{2\pi i(z-i)})}{(-2\pi)^{\lambda+1}} \{(s-1)_{(\lambda)} + i^k(k-s-1)_{(\lambda)}\}, \quad (56)$$

$$R_N(s, z) := \sum_{n=1}^{\infty} n^N e^{2\pi i n z} \left[ (s-1)_{(N+2)} \frac{\Gamma(s-N-2, -2\pi n)}{(-2\pi n)^s} + i^k(k-s-1)_{(N+2)} \frac{\Gamma(k-s-N-2, -2\pi n)}{(-2\pi n)^{k-s}} \right], \quad (57)$$

where  $(a)_{(\lambda)} := a(a-1)\cdots(a-\lambda+1)$  is the falling factorial ( $(a)_{(0)} := 1$ ),  $\text{Li}_s$  the polylogarithm of Definition 4.5, and  $\Gamma, \Gamma(a, x)$  the gamma and upper incomplete gamma, on principal branches [10].

**Lemma 4.13.** *Let  $f \in F_k$  have a pole of order  $p$  at  $z$  with projection weights  $r_{f,z}^*(m)$  (Definition 4.5). With  $\Pi_N, R_N$  as in (56)–(57), for  $\text{Im } z > 1$  and every  $s \in \mathbb{C}$ ,*

$$L^*(\hat{P}_z f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) = \sum_{m=1}^p r_{f,z}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}(s, z), \quad (58)$$

$$I_N(s, z) = \delta_{N,0} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} [\Pi_N(s, z) + R_N(s, z)], \quad (59)$$

and the second line's right side extends  $I_N(s, \cdot)$  continuously to  $\text{Im } z = 1$  for  $z \notin i + \mathbb{Z}$ .

The crux of this Lemma is to show the existence of the smooth continuation of  $I_N(s, z)$  from Lemma 4.12, derived for  $\text{Im}(z) > 1$ , to the  $\text{Im}(z) = 1$ -line away from the elliptic point  $z = i$ . This elliptic point requires special care, which must focus on, below.

*Proof.* We manipulate the main result of Lemma 4.12,

$$I_N(s, z) = \delta_{N,0} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n=1}^{\infty} n^N e^{2\pi i n z} \left( \frac{\Gamma(s, -2\pi n)}{(-2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi n)}{(-2\pi n)^{k-s}} \right), \quad (60)$$

Our proof proceeds in two steps. First, we isolate the possible sources of divergence on the line  $\text{Im}(z) = 1$  into a finite sum of polylogarithms  $\text{Li}_{\lambda+1-N}(e^{2\pi i(z-i)})$  for  $\lambda = 0, \dots, N+1$ . To do this, we use the recurrence relation [10] for each fixed- $n$  term,

$$\Gamma(a, x) = e^{-x} \sum_{\lambda=0}^{N+1} (a-1)_{(\lambda)} x^{a-1-\lambda} + (a-1)_{(N+2)} \Gamma(a-N-2, x) \quad (61)$$

to each gamma in  $G_{s,k}(-n)$  at  $x = -2\pi n$ . Dividing each incomplete-Gamma term by  $(-2\pi n)^s$  and  $(-2\pi n)^{k-s}$  yields, then, a common factor of  $1/(-2\pi n)^{\lambda+1}$  for each term in the finite sum. Further, this finite-sum is proportional to  $e^{2\pi n}$ ; we leave the trailing incomplete gammas as-is:

$$G_{s,k}(-n) = \left\{ e^{2\pi n} \sum_{\lambda=0}^{N+1} \frac{(s-1)_{(\lambda)}}{(-2\pi n)^{\lambda+1}} + \frac{(s-1)_{(N+2)} \Gamma(s-N-2, -2\pi n)}{(-2\pi n)^s} \right\} + i^k(s \rightarrow k-s). \quad (62)$$

Now, we consider the  $n$ -sum  $G_{s,k}(-n)$ , starting with the falling factorial terms:

$$\sum_{n=1}^{\infty} n^N e^{2\pi i n z} e^{2\pi n} \left( \sum_{\lambda=0}^{N+1} \frac{(s-1)_{(\lambda)}}{(-2\pi n)^{\lambda+1}} + i^k(s \rightarrow k-s) \right) = \sum_{\lambda=0}^{N+1} \frac{\text{Li}_{\lambda+1-N}(e^{2\pi i(z-i)})}{(-2\pi)^{\lambda+1}} \{(s-1)_{(\lambda)} + i^k(k-s-1)_{(\lambda)}\}, \quad (63)$$

i.e.,  $\Pi_N(s, z)$  of (56), while the reduced-gamma terms remaining are  $R_N(s, z)$  of Eq. (57), which together establish the lemma's identity (59).

The second main step of our proof is to show that  $\Pi_N(s, z)$  is regular as  $\text{Im}(z) \rightarrow 1$  for  $z \neq i$ , while  $R_N(s, z)$  is regular for all  $z$  with  $\text{Im}(z) = 1$ . We treat each piece in turn, beginning with  $R_N$ .

Each term of  $R_N$  is internal to the  $G_{s,k}(-n)$  in the mode  $\sum e^{2\pi i n z} G_{s,k}(-n)$  from Lemma 4.12. Clearly,  $|e^{2\pi i n z}| = e^{-2\pi n \text{Im} z}$ . Thus, the incomplete gamma's exponential growth cancels against the exponentially falling  $e^{2\pi i n z}$  factors. As  $n \rightarrow \infty$ , we have:

$$\left| n^N e^{2\pi i n z} \frac{\Gamma(s-N-2, -2\pi n)}{(-2\pi n)^s} \right| \sim \left| \frac{n^N e^{2\pi i n z} (-2\pi n)^{s-N-3} e^{2\pi n}}{(-2\pi n)^s} \right| \sim \left| \frac{e^{2\pi i n z} e^{2\pi n}}{(-2\pi)^{3+N} n^3} \right| \sim \frac{e^{2\pi n(1-\text{Im}(z))}}{(2\pi)^{N+3} n^3} \quad (64)$$

where the detailed asymptotics of  $\Gamma(s, x)$  in the first scaling relation follow from [10, §8.11]. (The  $i^k(s \rightarrow k-s)$ -partner follows identically.) Hence the  $n$ -th term of  $R_N$  is  $O(n^{-3} e^{-2\pi n(\text{Im} z - 1)})$ , uniformly for  $\text{Im} z \geq 1$ . There  $e^{-2\pi(\text{Im} z - 1)} \leq 1$ , so  $\sum_n n^{-3} e^{-2\pi n(\text{Im} z - 1)} \leq \zeta(3)$ : the series converges absolutely and uniformly, up to and including the edge  $\text{Im} z = 1$ , so  $R_N$  is continuous there.

For  $\Pi_N$ , recall that it is a sum over  $N+2$  polylogarithms of integer index,  $\Pi_N(s, z) = c_2(s) \text{Li}_2(w) + c_1(s) \text{Li}_1(w) + \dots + c_{1-N}(s) \text{Li}_{1-N}(w)$ , where  $w := e^{2\pi i(z-i)}$ . Each  $\text{Li}_a$  is single-valued and analytic on the cut plane  $\mathbb{C} \setminus [1, \infty)$ : for  $a \leq 0$  it is rational with a pole of order  $1-a$  at  $w = 1$ ;  $\text{Li}_1(w) = -\log(1-w)$  has a logarithmic branch point at  $w = 1$ ; and  $\text{Li}_2(w)$  is finite at its branch point  $w = 1$ . All three singularities sit at the endpoint  $w = 1$ , and the cuts of  $\text{Li}_1, \text{Li}_2$  run along the ray  $[1, \infty)$ . The edge is the unit circle  $|w| = 1$ , which meets that radial cut only at  $w = 1$ ; approaching it from inside the disc ( $\text{Im} z > 1$ ) gives the analytic boundary value of each  $\text{Li}_a$  at every  $w \neq 1$ . Hence  $\Pi_N$  is analytic along the whole edge except at  $z \in i + \mathbb{Z}$ , where each index  $\leq 1$  term diverges. With the continuity of  $R_N$ , (59) extends  $I_N(s, \cdot)$  continuously to  $\text{Im} z = 1$  for  $z \notin i + \mathbb{Z}$ .  $\square$

The one edge point the formula misses is  $z = i$ , where  $\Pi_N$  is singular ( $w = 1$ ); we compute  $I_N(s, i)$  directly. Its finiteness rests on a relation among the Laurent coefficients of  $f$  at the  $S$ -fixed point.

**Lemma 4.14.** *Let  $f \in F_k$  have a pole of order  $P$  at  $i$ , with pole weights  $r_{f,i}^*(m)$  (Definition 4.5), i.e. principal part  $\sum_{m=1}^P r_{f,i}^*(m)(\tau - i)^{-m}$ . Modularity  $f(-1/\tau) = \tau^k f(\tau)$  is equivalent to*

$$\sum_{m=r}^P r_{f,i}^*(m) \binom{m}{r} i^m = i^k \sum_{m=r}^P r_{f,i}^*(m) \binom{k}{m-r} i^{-m}, \quad r = 1, \dots, P. \quad (65)$$

The leading relation ( $r = P$ ) is  $((-1)^P - i^k) r_{f,i}^*(P) = 0$ , so the pole order matches  $k \bmod 4$  ( $P$  odd  $\Leftrightarrow k \equiv 2$ ,  $P$  even  $\Leftrightarrow 4 \mid k$ ); at  $P = 1$  this is  $(1 + i^k) r_{f,i}^*(1) = 0$ , and for a double pole the  $r = 1$  relation is  $r_{f,i}^*(1) = \frac{i(k-2)}{2} r_{f,i}^*(2)$ .

*Proof.* Write  $\tau = i + \varepsilon$ . Then  $-1/\tau - i = -\varepsilon/(1 - i\varepsilon)$ , so the principal part of  $f(-1/\tau)$  is  $\sum_m r_{f,i}^*(m)(i - \varepsilon^{-1})^m$  and that of  $\tau^k f(\tau)$  is  $i^k(1 - i\varepsilon)^k \sum_m r_{f,i}^*(m)\varepsilon^{-m}$ . Matching the coefficient of  $\varepsilon^{-r}$  through  $(i - \varepsilon^{-1})^m = \sum_j \binom{m}{j} i^{m-j} (-1)^j \varepsilon^{-j}$  and  $(1 - i\varepsilon)^k = \sum_l \binom{k}{l} (-i)^l \varepsilon^l$  gives  $(-1)^r \sum_{m=r}^P r_{f,i}^*(m) \binom{m}{r} i^{m-r} = i^k \sum_{m=r}^P r_{f,i}^*(m) \binom{k}{m-r} (-i)^{m-r}$ ; dividing through by  $i^r$  (with  $-1 = i^2$  and  $-i = i^{-1}$ ) yields (65).  $\square$

**Definition 4.15.** For  $N \geq 0$  and any cutoff  $J \geq N + 2$ , with  $H_N := \sum_{r=1}^N \frac{1}{r}$  and  $c_j^+(s) := (s-1)_{(j)} + i^k(k-s-1)_{(j)}$ , set

$$\mathcal{E}_N^+(s) := \sum_{n \geq 1} n^N e^{2\pi n} \left[ (s-1)_{(J)} \frac{\Gamma(s-J, 2\pi n)}{(2\pi n)^s} + i^k(k-s-1)_{(J)} \frac{\Gamma(k-s-J, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (66)$$

and then

$$\text{FP}_N(s, k) := \frac{c_N^+(s) H_N}{(2\pi)^{N+1}} + \sum_{\substack{0 \leq j \leq J \\ j \neq N}} \frac{c_j^+(s) \zeta(j+1-N)}{(2\pi)^{j+1}} + \frac{(-1)^N N!}{(2\pi)^{N+1}} \sum_{t=0}^{N-1} \frac{(-1)^t (s-1)_{(t)}}{t! (N-t)} + \mathcal{E}_N^+(s). \quad (67)$$

**Lemma 4.16.** Let  $f \in F_k$  have a pole of order  $P$  at  $i$  with pole weights  $r_{f,i}^*(m)$  (Definition 4.5), and  $\widehat{P}_i f$  its projection. Evaluated in the minimal class at basepoint  $i(1+\delta)$ , the pole's contribution to  $L^*(f, s)$  has the finite  $\delta \rightarrow 0$  limit

$$L^*(\widehat{P}_i f, s) := \lim_{\delta \rightarrow 0} L^*(\widehat{P}_i f, s; i(1+\delta)) = \sum_{m=1}^P \frac{(-2\pi i)^m r_{f,i}^*(m)}{(m-1)!} \text{FP}_{m-1}(s, k), \quad (68)$$

entire in  $s$ . The limit exists precisely because the pole weights obey the relations of Lemma 4.14 at the  $S$ -fixed point  $i$ : they cancel the individually divergent contour pieces.

*Proof.* Fix  $s$  and evaluate in the minimal class at basepoint  $i(1+\delta)$ ,  $\delta > 0$ ; we show the  $\delta \rightarrow 0$  limit exists and equals the right side.

On this contour the pole at  $i$  lies strictly *below* the  $T$ -segment (height  $1+\delta$ ) and *on* the  $S$ -segment, which is taken by principal value. Because the pole is below the  $T$ -segment,  $|e^{2\pi i(\tau-i)}| < 1$  there and each block  $\text{Li}_{1-m}(e^{2\pi i(\tau-i)}) = \sum_{n \geq 1} n^{m-1} e^{2\pi i n(\tau-i)}$  is used *without* inversion. In particular no constant ( $n=0$ ) mode is generated, so no term  $\frac{1}{s} + \frac{i^k}{k-s}$  arises:  $L^*(\widehat{P}_i f, s)$  is entire in  $s$ . (Contrast a pole *above* the contour, where Lemma 4.12 inverts  $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$  and the leftover  $-1$  produces exactly that pole; here it is absent.)

*Finite parts assemble to FP.* Expanding block  $m$  about  $i$  and integrating term-by-term — against  $\widetilde{k}_T$  on the  $T$ -segment and against  $\tau^{s-1}$  on the  $S$ -segment — its finite part is  $\frac{(-2\pi i)^m}{(m-1)!} r_{f,i}^*(m) \text{FP}_{m-1}(s, k)$ , with the harmonic term  $c_{m-1}^+(s) H_{m-1}$ , the  $\zeta$ -sum, the polynomial term, and the convergent remainder  $\mathcal{E}_{m-1}^+(s)$  of Definition 4.15 arising as the  $\tau \rightarrow i$  expansion coefficients of the block (the remainder as in Lemma 4.13).

*Divergences cancel by Lemma 4.14.* Beyond its finite part, block  $m$  throws off a divergence  $\sim \delta^{-(m-1)}, \dots, \log \delta$  from the  $S$ -segment principal value at the endpoints straddling  $i$ . Collected at each order  $\delta^{-r}$ , the coefficient is a fixed linear combination of the weights  $\sum_m(\cdot) r_{f,i}^*(m)$  — precisely the combination that the modularity relations (65) set to zero at the  $S$ -fixed point. Every  $\delta^{-r}$  and  $\log \delta$  term therefore vanishes and the limit is finite; were the weights to violate (65), it would diverge logarithmically.

Taking  $\delta \rightarrow 0$  leaves the FP-sum (68). The cutoff  $J$  is immaterial: raising it moves terms between the  $\zeta$ -sum and  $\mathcal{E}_N^+$ , which stays convergent as in Lemma 4.13.  $\square$

**Theorem 4.17.** Let  $f \in F_k$  have poles  $\tau_p$  of order  $P_p$  with  $\text{Im } \tau_p \geq 1$  in the strip  $|\text{Re } \tau| \leq \frac{1}{2}$ , pole-projected Fourier coefficients  $c_{Rf}(n)$  (Definition 4.7), and projection weights  $r_{f,\tau_p}^*(m)$  (Defini-

tion 4.5). In the minimal class,

$$L^*(f, s) = -c_{Rf}(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n) + \sum_{\tau_p \neq i} \sum_{m=1}^{P_p} r_{f, \tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}(s, \tau_p) + L^*(\hat{P}_i f, s), \quad (69)$$

with  $I_{m-1}$  from Lemmas 4.12–4.13 and the last term (present only for a pole at  $i$ ) from Lemma 4.16. The right side is entire in  $s$  apart from at most simple poles at  $s = 0$  and  $s = k$ , and reduces to the weakly-holomorphic  $L^*$  of Theorem 3.2 when  $f$  is holomorphic on  $\mathbb{H}$ .

*Proof.* Split  $f = (I - \hat{P})f + \hat{P}f$ . By Lemma 4.10 the Fourier series of the pole-projected part converges on  $\hat{\gamma}_f^T$ ; Theorem 3.2 applied term by term gives its  $L^*$  as the first two sums. The projection  $\hat{P}f = \sum_{\tau_p} \sum_m r_{f, \tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$  contributes, for each  $\tau_p \neq i$ , the  $L$ -integrals  $I_{m-1}(s, \tau_p)$  of Definition 4.11, evaluated by Lemmas 4.12–4.13; the pole at  $i$  contributes  $L^*(\hat{P}_i f, s)$  of Lemma 4.16. Entirety and the pole locations follow from those of  $G_{s,k}$  and the  $\delta_{N,0}$  terms; when  $f$  is holomorphic on  $\mathbb{H}$ ,  $\hat{P}f = 0$  and  $c_{Rf} = c_f$ , recovering Theorem 3.2.  $\square$

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